

A First Course in Causal Inference Ch.5

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Ch.5 Stratification and Post-Stratification in Randomized Experiments

Ch.5 Stratification and Post-Stratification in Randomized Experiments

This chapter explains the meaning of the following quote:

Block what you can and randomize what you cannot.

– *Box et al. (1978)*

5.1 Stratification

- ▶ A CRE may generate an undesired treatment allocation by chance.
- ▶ Setup: a CRE with a discrete covariate $X_i \in \{1, \dots, K\}$; define $n_{[k]} = \#\{i: X_i = k\}$ and $\pi_{[k]} = n_{[k]}/n$ as the number and proportion of units in stratum k ($k = 1, \dots, K$).
- ▶ Then, $n_{[k]1} = \#\{i: X_i = k, Z_i = 1\}$ and $n_{[k]0} = \#\{i: X_i = k, Z_i = 0\}$.
- ▶ With positive probability, $n_{[k]1}$ or $n_{[k]0}$ is zero for some k : **it is possible that some strata only have treated or control units.**
- ▶ So, the proportions of units in stratum k are different across the treatment and control groups although on average their difference is zero:

$$\mathbb{E} \left[\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0} \right] = 0. \quad (1)$$

5.1 Stratification

Proof for Eq. (1).

We also write $n_{[k]1}$ and $n_{[k]0}$ as $n_{[k]1} = \sum_{i: X_i=k} Z_i$ and $n_{[k]0} = \sum_{i: X_i=k} (1 - Z_i)$, respectively. We then obtain

$$\mathbb{E} \left[\frac{n_{[k]1}}{n_1} \right] = \frac{1}{n_1} \mathbb{E}[n_{[k]1}] = \frac{1}{n_1} \mathbb{E} \left[\sum_{i: X_i=k} Z_i \right] = \frac{1}{n_1} \sum_{i: X_i=k} \mathbb{E}[Z_i] = \frac{1}{n_1} n_{[k]} \frac{n_1}{n} = \frac{n_{[k]}}{n}.$$

Similarly,

$$\mathbb{E} \left[\frac{n_{[k]0}}{n_0} \right] = \frac{n_{[k]}}{n}.$$

Hence, $\mathbb{E} \left[\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0} \right] = 0. \quad \square$

5.1 Stratification

- ▶ When $\frac{n_{[k]1}}{n_1} - \frac{n_{[k]0}}{n_0}$ is large for some strata k 's, the treatment and control groups have undesirable covariate imbalance.
- ▶ In an extreme example, suppose $k = 2$ and $n = 8$, $n_1 = 4$, $n_0 = 4$, $n_{[1]} = 4$, and $n_{[2]} = 4$. There exists a case $n_{[1]1} = 4$, $n_{[1]0} = 0$, $n_{[2]1} = 0$, and $n_{[2]0} = 4$. Here there is undesirable covariate imbalance amongst both groups.
- ▶ To avoid such covariate imbalance in experiments, we introduce *stratified randomized experiments* (SRE).

5.1 Stratification

Definition 5.1 (SRE)

Fix the $n_{[k]1}$'s or $n_{[k]0}$'s. We conduct K independent CREs within the K strata of a discrete covariate X .

- ▶ The total number of randomizations in an SRE equals

$$\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}},$$

and each feasible randomization has equal probability.

- ▶ Within stratum k , the proportion of units receiving the treatment is

$$e_{[k]} = \frac{n_{[k]1}}{n_{[k]}},$$

which is also called the *propensity score*.

5.1 Stratification

An SRE is different from a CRE:

1.

$$\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}} < \binom{n}{n_1}.$$

► In the previous example, in a CRE the number of allocation is $\binom{8}{4} = 70$

while that of allocation is $\binom{4}{2} \binom{4}{2} = 36$. Hence $70 > 36$.

2. $e_{[k]}$ is fixed in an SRE but random in a CRE

► It says the exactly same thing to the previous illustration.

5.1 Stratification

- For stratum k , we have the stratum-specific average causal effect

$$\tau_{[k]} = \frac{1}{n_{[k]}} \sum_{X_i=k} \tau_i.$$

- The average causal effect is¹

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \frac{1}{n} \sum_{k=1}^K \sum_{X_i=k} \tau_i = \frac{1}{n} \sum_{k=1}^K n_{[k]} \tau_{[k]} = \sum_{k=1}^K \frac{n_{[k]}}{n} \tau_{[k]} = \sum_{k=1}^K \pi_{[k]} \tau_{[k]}.$$

¹The third equality follows from the equation above: $\sum_{X_i=k} \tau_i = n_{[k]} \tau_{[k]}$.

5.2 FRT

- ▶ The FRT in an SRE.
- ▶ The sharp null hypothesis: $H_{0F} : Y_i(1) = Y_i(0)$ for all units $i = 1, \dots, n$.
- ▶ Following the Def. 5.1, when we simulate the treatment vector, we must permute the treatment indicators within strata of X .
- ▶ We should choose test statistics that can reflect the nature of the SRE.

5.2 FRT

Example 5.1 Stratified estimator.

Motivated by estimating τ , we can use the following stratified estimator in the FRT:

$$\hat{\tau}_S = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]},$$

where

$$\hat{\tau}_{[k]} = \frac{1}{n_{[k]1}} \sum_{i=1}^n \mathbb{I}(X_i = k, Z_i = 1) Y_i - \frac{1}{n_{[k]0}} \sum_{i=1}^n \mathbb{I}(X_i = k, Z_i = 0) Y_i$$

is the stratum-specific difference-in-means within stratum k .

5.2 FRT

Example 5.2 Studentized stratified estimator.

Motivated by the studentized statistic in the simple two-sample problem, we can use the following studentized statistic for the stratified estimator in the FRT:

$$t_S = \frac{\hat{\tau}_S}{\sqrt{\hat{V}_S}},$$

with

$$\hat{V}_S = \sum_{k=1}^K \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right)$$

where $\hat{S}_{[k]}^2(1)$ and $\hat{S}_{[k]}^2(0)$ are the stratum-specific sample variances of the outcomes under the treatment and control, respectively.

5.2 FRT

Example 5.3 Combining Wilcoxon rank-sum statistics.

We first compute the Wilcoxon rank sum statistic $W_{[k]}$ ² within stratum k and then combine them as

$$W_S = \sum_{k=1}^K c_{[k]} W_{[k]}.$$

For $c_{[k]}$, there are two proposed methods:

$$c_{[k]} = \frac{1}{n_{[k]1} n_{[k]0}},$$
$$c_{[k]} = \frac{1}{n_{[k]} + 1}.$$

²Recall the Wilcoxon rank sum statistic is defined as $W = \sum_{i=1}^n Z_i R_i$, where $R_i = \#\{j: Y_j \leq Y_i\}$.

5.2 FRT

Example 5.4 Hodges Jr and Lehmann (2011)'s aligned rank statistics. Hodges Jr and Lehmann (2011) proposed a test statistic that make more comparisons across strata after standardizing the outcomes. First,

$$\tilde{Y}_i = Y_i - \bar{Y}_{[k]}$$

with $\bar{Y}_{[k]} = \frac{1}{n_{[k]}} \sum_{X_i=k} Y_i$, then obtaining the ranks $(\tilde{R}_1, \dots, \tilde{R}_n)$ of the pooled outcomes $(\tilde{Y}_1, \dots, \tilde{Y}_n)$, and finally constructing the test statistic

$$\tilde{W} = \sum_{i=1}^n Z_i \tilde{R}_i.$$

5.3 Neymanian inference

- ▶ Statistical inference for an SRE builds on the fact that it essentially consists of K independent CREs.
- ▶ Within stratum k , the difference-in-means $\hat{\tau}_{[k]}$ is unbiased for $\tau_{[k]}$ with variances

$$\text{var}(\hat{\tau}_{[k]}) = \frac{S_{[k]}^2(1)}{n_{[k]1}} + \frac{S_{[k]}^2(0)}{n_{[k]0}} - \frac{S_{[k]}^2(\tau)}{n_{[k]}}.$$

- ▶ Hence, the stratified estimator $\hat{\tau}_S = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]}$ is unbiased for $\tau = \sum_{k=1}^K \pi_{[k]} \tau_{[k]}$ with variance

$$\text{var}(\hat{\tau}_S) = \sum_{k=1}^K \pi_{[k]}^2 \text{var}(\hat{\tau}_{[k]}).$$

5.3 Neymanian inference

- ▶ Similarly, we obtain a conservative variance estimator

$$\hat{V}_S = \sum_{k=1}^K \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right),$$

with sample variances of the outcomes $\hat{S}_{[k]}^2(1)$ and $\hat{S}_{[k]}^2(0)$.

- ▶ We can also construct a Wald-type $1 - \alpha$ CI for τ and p -values.
- ▶ In short, **SRE is essentially K independent CREs.**

5.3 Neymanian inference

- ▶ What are the benefits of the SRE compared with the CRE?
- ▶ Benefits from the covariate balance in the SRE.
- ▶ Better covariate balance results in better estimation precision of the average causal effect.

5.3 Neymanian inference

Assume that $e_{[k]} = e$ for all k . Then

$$\hat{\tau} = \hat{\tau}_S. \quad (2)$$

► Proof

5.3 Neymanian inference

To compare $\text{var}_{CRE}(\hat{\tau})$ and $\text{var}_{SRE}(\hat{\tau}_S)$, we first derive $S^2(1)$, $S^2(0)$ and $S^2(\tau)$. We then obtain

$$\begin{aligned} S^2(1) &= \sum_{k=1}^K \left[\frac{n_{[k]} - 1}{n - 1} S_{[k]}^2(1) + \frac{n_{[k]}}{n - 1} \{ \bar{Y}_{[k]}(1) - \bar{Y}(1) \}^2 \right], \\ S^2(0) &= \sum_{k=1}^K \left[\frac{n_{[k]} - 1}{n - 1} S_{[k]}^2(0) + \frac{n_{[k]}}{n - 1} \{ \bar{Y}_{[k]}(0) - \bar{Y}(0) \}^2 \right], \\ S^2(\tau) &= \sum_{k=1}^K \left[\frac{n_{[k]} - 1}{n - 1} S_{[k]}^2(\tau) + \frac{n_{[k]}}{n - 1} \{ \tau_{[k]} - \tau \}^2 \right]. \end{aligned}$$

5.3 Neymanian inference

Hence,

$$\begin{aligned} & \text{var}_{CRE}(\hat{\tau}) \\ &= \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} \\ &= \sum_{k=1}^K \left[\frac{n_{[k]} - 1}{(n-1)n_1} S_{[k]}^2(1) + \frac{n_{[k]} - 1}{(n-1)n_0} S_{[k]}^2(0) - \frac{n_{[k]} - 1}{(n-1)n} S_{[k]}^2(\tau) \right] \\ &+ \sum_{k=1}^K \left[\frac{n_{[k]}}{(n-1)n_1} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 + \frac{n_{[k]}}{(n-1)n_0} \{\bar{Y}_{[k]}(0) - \bar{Y}(0)\}^2 - \frac{n_{[k]}}{(n-1)n} \{\tau_{[k]} - \tau\}^2 \right] \end{aligned}$$

5.3 Neymanian inference

With large $n_{[k]}$'s, $n_{[k]} - 1 \approx n_{[k]}$ and $n - 1 \approx n$ then

$$\begin{aligned}\text{var}_{CRE}(\hat{\tau}) &\approx \sum_{k=1}^K \left[\frac{\pi_{[k]}}{n_1} S_{[k]}^2(1) + \frac{\pi_{[k]}}{n_0} S_{[k]}^2(0) - \frac{\pi_{[k]}}{n} S_{[k]}^2(\tau) \right] \\ &\quad + \sum_{k=1}^K \left[\frac{\pi_{[k]}}{n_1} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 + \frac{\pi_{[k]}}{n_0} \{\bar{Y}_{[k]}(0) - \bar{Y}(0)\}^2 - \frac{\pi_{[k]}}{n} \{\tau_{[k]} - \tau\}^2 \right].\end{aligned}$$

Following the assumption $e_{[k]} = e$, we write $\text{var}_{SRE}(\hat{\tau})$ as

$$\begin{aligned}\text{var}_{SRE}(\hat{\tau}_S) &= \sum_{k=1}^K \pi_{[k]}^2 \left[\frac{S_{[k]}^2(1)}{n_{[k]1}} + \frac{S_{[k]}^2(0)}{n_{[k]0}} - \frac{S_{[k]}^2(\tau)}{n_{[k]}} \right] \\ &= \sum_{k=1}^K \left[\frac{\pi_{[k]}}{n_1} S_{[k]}^2(1) + \frac{\pi_{[k]}}{n_0} S_{[k]}^2(0) - \frac{\pi_{[k]}}{n} S_{[k]}^2(\tau) \right].\end{aligned}$$

5.3 Neymanian inference

With large $n_{[k]}$'s, approximately,

$$\begin{aligned} & \text{var}_{CRE}(\hat{\tau}) - \text{var}_{SRE}(\hat{\tau}_S) \\ & \approx \sum_{k=1}^K \left[\frac{\pi_{[k]}}{n_1} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 + \frac{\pi_{[k]}}{n_0} \{\bar{Y}_{[k]}(0) - \bar{Y}(0)\}^2 - \frac{\pi_{[k]}}{n} \{\tau_{[k]} - \tau\}^2 \right] \\ & \geq 0. \end{aligned} \tag{3}$$

The equality holds only in a case that for $k = 1, \dots, K$,

$$\sqrt{\frac{n_0}{n_1}} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 + \sqrt{\frac{n_1}{n_0}} \{\bar{Y}_{[k]}(0) - \bar{Y}(0)\} = 0.$$

5.3 Neymanian inference

This result has several implications:

- ▶ An SRE is generally more efficient than a CRE, especially when the stratifying covariate is predictive of the potential outcomes.
- ▶ In the extreme case where the stratifying covariate contains little information about the potential outcomes (that is, the potential outcomes are nearly the same across strata), an SRE provides little or no efficiency gain over a CRE.
- ▶ This comparison relies on the large-strata approximation.
- ▶ Therefore, the result does not necessarily hold in finite samples; in some cases, an SRE can even be less efficient than a CRE.

5.4 Post-stratification in a CRE

Let $\mathbf{n} = \{n_{[k]1}, n_{[k]0}\}_{k=1}^K$, then a CRE becomes an SRE, because

$$\text{pr}_{CRE}(\mathbf{Z} = \mathbf{z} \mid \mathbf{n}) = \frac{\text{pr}_{CRE}(\mathbf{Z} = \mathbf{z}, \mathbf{n})}{\text{pr}_{CRE}(\mathbf{n})} = \frac{1}{\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}}}. \quad (4)$$

► Proof

Eq. (4) means that the conditional distribution of $\mathbf{Z}$ from a CRE given $\mathbf{n}$ is identical to the distribution of $\mathbf{Z}$ from an SRE. Hence, the Neymanian analysis becomes *post-stratification*:

$$\hat{\tau}_{PS} = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]},$$

which has an identical form as $\hat{\tau}_S$.

5.4 Post-stratification in a CRE

- ▶ Post-stratification improves efficiency compared to $\hat{\tau}$, if
 - ▶ Covariates X is predictive of the outcomes,
 - ▶ and this holds under a limited number of data-generating processes and under that all strata are large enough.
- ▶ Larger K results in some $n_{[k]1} = 0$ or some $n_{[k]0} = 0$
 $\rightsquigarrow$ small or zero values of $n_{[k]1}$ or $n_{[k]0}$ possibly reduce the power.
- ▶ Stratification uses X in the design stage and post-stratification uses X in the analysis stage.

Proof of Eq. (2)

Recall

$$\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i - \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i, \quad \hat{\tau}_S = \sum_{k=1}^K \pi_{[k]} \hat{\tau}_{[k]},$$

where

$$\hat{\tau}_{[k]} = \frac{1}{n_{[k]1}} \sum_{i: X_i=k} Z_i Y_i - \frac{1}{n_{[k]0}} \sum_{i: X_i=k} (1 - Z_i) Y_i, \quad \pi_{[k]} = \frac{n_{[k]}}{n}$$

Then,

$$\begin{aligned} \hat{\tau}_S &= \sum_{k=1}^K \frac{1}{n \cdot \frac{n_{[k]1}}{n_{[k]}}} \sum_{i: X_i=k} Z_i Y_i - \sum_{k=1}^K \frac{1}{n \cdot \frac{n_{[k]0}}{n_{[k]}}} \sum_{i: X_i=k} (1 - Z_i) Y_i \\ &= \frac{1}{n_1} \sum_{k=1}^K \sum_{i: X_i=k} Z_i Y_i - \frac{1}{n_0} \sum_{k=1}^K \sum_{i: X_i=k} (1 - Z_i) Y_i = \hat{\tau}. \quad \square \end{aligned}$$

The second equality follows from the assumption $e = e_{[k]} = n_{[k]1}/n_{[k]}$.

Variances Derivation

$$\begin{aligned} S^2(1) &= \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2 \\ &= \frac{1}{n-1} \sum_{k=1}^K \sum_{i: X_i=k} \{Y_i(1) - \bar{Y}_{[k]}(1) + \bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 \\ &= \frac{1}{n-1} \sum_{k=1}^K \sum_{i: X_i=k} [\{Y_i(1) - \bar{Y}_{[k]}(1)\}^2 + \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2] \\ &= \sum_{k=1}^K \left[\frac{n_{[k]} - 1}{n-1} S_{[k]}^2(1) + \frac{n_{[k]}}{n-1} \{\bar{Y}_{[k]}(1) - \bar{Y}(1)\}^2 \right]. \quad \square \end{aligned}$$

Proof of Eq. (3)

Let $A_k = \bar{Y}_{[k]}(1) - \bar{Y}()1$ and $B_k = \bar{Y}_{[k]}(0) - \bar{Y}()0$. Because $\tau_{[k]} = \bar{Y}_{[k]}(1) - \bar{Y}_{[k]}(0)$ and $\tau = \bar{Y}(1) - \bar{Y}(0)$, we write the difference as:

$$\begin{aligned} \sum_{k=1}^K \left[\frac{\pi_{[k]}}{n_1} A_k^2 + \frac{\pi_{[k]}}{n_0} B_k^2 - \frac{\pi_{[k]}}{n} (A_k - B_k)^2 \right] &= \sum_{k=1}^K \frac{\pi_{[k]}}{n} \left[\frac{n}{n_1} A_k^2 + \frac{n}{n_0} B_k^2 - (A_k - B_k)^2 \right] \\ &= \sum_{k=1}^K \frac{\pi_{[k]}}{n} \left[\left(\frac{n}{n_1} - 1 \right) A_k^2 + \left(\frac{n}{n_0} - 1 \right) B_k^2 + 2A_k B_k \right] \\ &= \sum_{k=1}^K \frac{\pi_{[k]}}{n} \left[\frac{n_0}{n_1} A_k^2 + \frac{n_1}{n_0} B_k^2 + 2A_k B_k \right] \\ &= \sum_{k=1}^K \frac{\pi_{[k]}}{n} \left\{ \sqrt{\frac{n_0}{n_1}} A_k + \sqrt{\frac{n_1}{n_0}} B_k \right\}^2 \geq 0. \quad \square \end{aligned}$$

Proof of Eq. (4)

In a CRE, $\text{pr}_{CRE}(\mathbf{Z} = \mathbf{z}) = 1 / \binom{n}{n_1}$ for every $\mathbf{z}$ with $\sum_i z_i = n_1$.

Building on Sec. 5.1, the total number of randomizations in an SRE is $\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}}$, and these form a subset of all assignments feasible in a CRE. Hence

$$\text{pr}_{CRE}(\mathbf{n}) = \frac{\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}}}{\binom{n}{n_1}}.$$

If $\mathbf{z}$ is compatible with $\mathbf{n}$, $\text{pr}_{CRE}(\mathbf{Z} = \mathbf{z}, \mathbf{n}) = \text{pr}_{CRE}(\mathbf{Z} = \mathbf{z})$. Therefore

$$\begin{aligned} \text{pr}_{CRE}(\mathbf{Z} = \mathbf{z} \mid \mathbf{n}) &= \frac{\text{pr}_{CRE}(\mathbf{Z} = \mathbf{z}, \mathbf{n})}{\text{pr}_{CRE}(\mathbf{n})} = \frac{1}{\binom{n}{n_1}} \cdot \frac{\binom{n}{n_1}}{\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}}} \\ &= \frac{1}{\prod_{k=1}^K \binom{n_{[k]}}{n_{[k]1}}}. \quad \square \end{aligned}$$

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- Hodges Jr, J. L. and E. L. Lehmann (2011). Rank methods for combination of independent experiments in analysis of variance. In *Selected works of EL Lehmann*, pp. 403–418. Springer.